

SATELLITE IMAGE SEGMENTATION WITH PRETRAINED MODELS

Pretraining on satellite imagery for data-scarce land-cover segmentation

Neural Networks - A.Y. 2025/2026
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PROJECT AIMS

GOAL

- demonstrate that domain-specific remote sensing pre-training overcomes data scarcity, outperforming models trained from scratch
- test our model with Multispectral Bands Imagery Data

SCRATCH

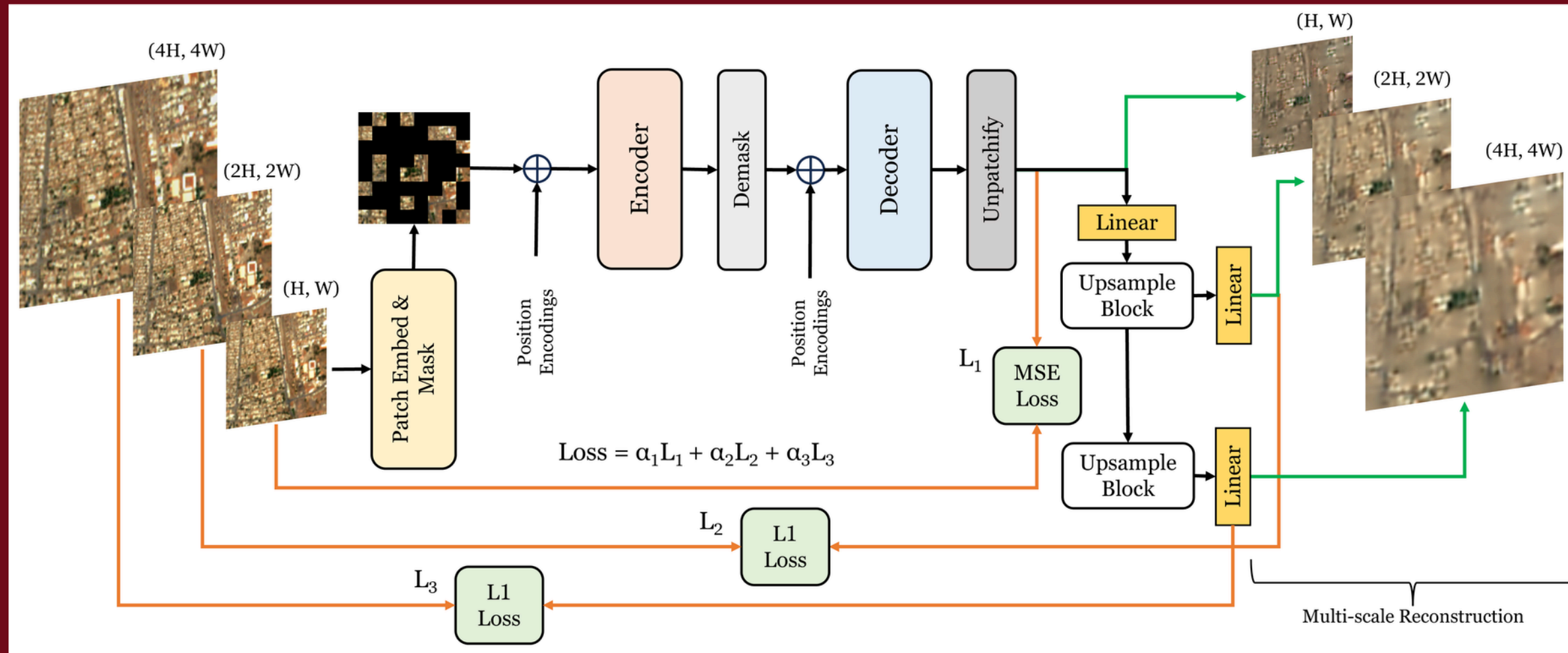
VS.

**PRE-TRAINED
WEIGHTS**



MATERIAL OVERVIEW

- Rethinking Transformers Pre-training for Multi-Spectral Satellite Imagery



- <https://github.com/ViTAE-Transformer/RSP/tree/main>
- https://github.com/techmn/satmae_pp

IMPLEMENTATION

DATASET AND PARAMETERS

- LoveDA - 7 classes (background, building, road, water, barren, forest, agriculture)
 - RGB images
- DFC2020 - Multispectral and SAR data (Sentinel-2)
 - 8 classes
- implementations: batch size of 32, 15/20 epochs, resize and crop images to 224x224 patches, class-weighting, binary mode
- data scarcity scenario represented by tiny subset of training dataset

Two Architectures:

- Swin-T U-Net

Pre-training: RSP (Remote Sensing Pre-training on *MillionAD*)

- ViT-Large

Pre-training: SatMAE++ (Group-channel masking on *fMoW*)



COMPARISON (Swin U-Net + RSP)

Train Swin U-Net network first completely from scratch, then with the RSP weights.

```
=====
RUN: swin_noweights | pretraining=scratch | train_subset=300 | epochs=20
Device: cuda | img_size=224 | batch=32 | Wavelet: False | class_weights=True | ignore_index=0
=====
Input mode: crop (crop 224 da resize 512)
Initializing TorchGeo LOVEDA (train split)...
Files already downloaded and verified
Initializing TorchGeo LOVEDA (val split)...
Files already downloaded and verified
Train samples: 300 | Val samples: 400
Initializing U-Net | backbone: tu-swin_tiny_patch4_window7_224 | pretraining: scratch
Backbone smp caricato.
○ Backbone RANDOM (baseline: nessun pretraining).
Class-weighting ATTIVO: [0.0, 0.21, 1.04, 2.29, 1.22, 1.0, 0.57, 0.3]
```

best mIoU = 0.0854



deep difference

```
=====
RUN: swin_rsp | pretraining=rsp | train_subset=300 | epochs=20
Device: cuda | img_size=224 | batch=32 | Wavelet: False | class_weights=True | ignore_index=0
=====
Input mode: crop (crop 224 da resize 512)
Initializing TorchGeo LOVEDA (train split)...
Files already downloaded and verified
Initializing TorchGeo LOVEDA (val split)...
Files already downloaded and verified
Train samples: 300 | Val samples: 400
Initializing U-Net | backbone: tu-swin_tiny_patch4_window7_224 | pretraining: rsp
Backbone smp caricato.
Pesi satellitari Swin-T (RSP) iniettati con successo!
Class-weighting ATTIVO: [0.0, 0.21, 1.04, 2.29, 1.22, 1.0, 0.57, 0.3]
Backbone congelato per i primi 3 epoch (warmup)...
```

best mIoU = 0.2590



COMPARISON (ViT + SatMAE++)

Test a Vision Transformer from scratch against the version initialized with SatMAE++ weights

```
=====
RUN: vit_noweights | pretraining=satmaepp_rand | train_subset=300 | epochs=20
Device: cuda | img_size=224 | batch=32 | Wavelet: False | class_weights=True | ignore_index=0
=====
Input mode: crop (crop 224 da resize 512)
Initializing TorchGeo LOVEDA (train split)...
Files already downloaded and verified
Initializing TorchGeo LOVEDA (val split)...
Files already downloaded and verified
Train samples: 300 | Val samples: 400
[SatMAE++] Encoder ViT-L RANDOM (controllo: stessa rete SENZA pretraining).
Class-weighting ATTIVO: [0.0, 0.21, 1.04, 2.29, 1.22, 1.0, 0.57, 0.3]
```

best mIoU = 0.1690



deep difference

```
=====
RUN: vit_satmae | pretraining=satmaepp | train_subset=300 | epochs=20
Device: cuda | img_size=224 | batch=32 | Wavelet: False | class_weights=True | ignore_index=0
=====
Input mode: crop (crop 224 da resize 512)
Initializing TorchGeo LOVEDA (train split)...
Files already downloaded and verified
Initializing TorchGeo LOVEDA (val split)...
Files already downloaded and verified
Train samples: 300 | Val samples: 400
[SatMAE++] Pesi caricati: 294/294 tensori (missing=0, unexpected=16).
[SatMAE++] Encoder ViT-L inizializzato con pesi SatMAE++ (congelato).
Class-weighting ATTIVO: [0.0, 0.21, 1.04, 2.29, 1.22, 1.0, 0.57, 0.3]
```

best mIoU = 0.3770



COMPARISON (ViT + MSI)

Scenario: a) MSI (no weights)

b) MSI (with weights)

c) MSI + SAR fusion (with weights)

```
model=satmae | bands=msi | binary=False | classes=8 | train 2000 | val 800 | dev=cuda
[SatMAE-S] checkpoint non trovato -> encoder RANDOM (controllo)
class-weights: [0.69, 10.0, 0.92, 10.0, 0.63, 1.09, 10.0, 0.49]
```

```
model=satmae | bands=msi | binary=False | classes=8 | train 2000 | val 800 | dev=cuda
[SatMAE-S] pesi 299/302 (missing=3, unexpected=24)
class-weights: [0.69, 10.0, 0.92, 10.0, 0.63, 1.09, 10.0, 0.49]
```

```
model=satmae | bands=msi_sar | binary=False | classes=8 | train 2000 | val 800 | dev=cuda
[SatMAE-S] pesi 299/302 (missing=3, unexpected=24)
class-weights: [0.69, 10.0, 0.92, 10.0, 0.63, 1.09, 10.0, 0.49]
```

a) $mIoU = 0.4799$



slight difference



b) $mIoU = 0.5210$

c) $mIoU = 0.5346$



slight variation due to speckle noise and
stochastic nature of the training



CONCLUSIONS & FUTURE WORKS

CONCLUSIONS

In Remote Sensing, where labeled data is scarce and expensive, training from scratch is not a viable path.

Domain-specific pre-training (RSP, SatMAE++) and multimodal fusion (integrating radar) are necessary conditions for obtaining models that are stable, fast to converge, and accurate.

LIMITATIONS

Due to the hardware restrictions of free GPUs, we had to drastically reduce the amount of images used for training. Consequently, the absolute accuracy and mIoU reached are not very high, since Transformers are extremely 'data-hungry' models to perform at their best.

However, the differences between the models prove weights effectiveness.

FUTURE WORKS

- train the model on a very larger dataset to unlock the network's true performance
- extend the multimodal fusion by integrating it with time-series analysis (images taken on different dates) to monitor land evolution over time

